

Writing Two-Step Equations from Real-World Contexts

Worksheet 7A.01.06

Name _____

Date _____

Period _____

For questions 1 and 2, name the known quantities and the unknown quantity. Then write an equation. Do not solve.

1. A town's water tower held 110,000 gallons and is draining at a rate of 2,150 gallons per hour. It now holds 79,900 gallons. How long has it been draining?

Known quantities.

Unknown quantity.

Equation.

2. A giraffe born at a local zoo was 6 feet tall at birth and grows 0.08 foot each day during its first week. It is now 6.48 feet tall. How many days old is it?

Known quantities.

Unknown quantity.

Equation.

For questions 3 through 6, write an equation, solve it, and write one sentence saying what your solution means.

3. The average toddler grows 0.2 inch per month. James is 35.45 inches tall. If he grows at the average rate, how long will it take him to reach a height of 37 inches?

- Mr. Humphries has some quarters and four dimes in his pocket, worth \$2.40 in all. How many quarters does he have?
- A phone plan costs \$24.99 a month plus \$0.12 for each gigabyte of data used. Last month's bill was \$32.19. How many gigabytes were used?
- Four concert tickets were purchased. A \$2.50 service fee was added to each ticket. The total came to \$118.00. What was the price of one ticket before the fee?

Questions 7 and 8 involve percentages. Read the order of the steps carefully.

7. Maddox has a coupon for \$10 off his purchase at Foot Zone. He found a pair of shoes marked 20% off. The store takes the 20% off first and then applies the coupon. Maddox paid \$36.20 before taxes. What was the original price of the shoes?
8. Victoria works as a salesperson. She earns \$350 each week plus a 10% commission on her weekly sales. This week her total earnings need to be \$800. How much does she need to sell?

Questions 9 and 10 look ahead to multi-step equations, which arrive later this year. In each one the unknown appears on both sides of the equal sign. Write the equation first, then solve.

- 9.** Gym A charges \$30 to join plus \$18 a month. Gym B charges \$75 to join plus \$12 a month. After how many months does a membership cost the same at both gyms?

- 10.** Plan A costs \$45 a month plus \$0.10 for each gigabyte of data. Plan B costs \$30 a month plus \$0.25 for each gigabyte. At how many gigabytes do the two plans cost the same?